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JEE MAIN

REVIEW TEST-5

CODE : 20105

Class : XI-JEE

Date : 22-08-2026

Time 3 Hours.

Max. Marks : 300

IMPORTANT INSTRUCTIONS

1. The question paper consists of 3 Parts (Part-1 contains Maths, Part-2 contains Physics and Part-3 contains Chemistry. Please fill the OMR Answer Sheet accordingly and carefully.
2. Each part contains (20 questions in section-1 and 5 questions in Section-2)
3. Please ensure that the Question Paper you have received contains ALL THE QUESTIONS in each Section and PAGES. If you find any mistake like missing questions or pages then contact immediately.
4. Use of **Calculator, Log Table, Slide Rule and Mobile** is not allowed.

:- MARKING SCHEME :-

SECTION : 1

- This section contains **TWENTY** (20) questions.
- Each question has **FOUR** options. **ONLY ONE** of these four option is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme :**

Full Mark : +4. If **ONLY** the correct option is chosen;

Zero Marks : 0. If none of the option is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

SECTION : 2

- This section contains **FIVE** (05) questions. The answer to each question is **INTEGER TYPE**.
 - If answer is in **Decimal then Round off to Nearest Integer**.
 - Answer to each question will be evaluated **according to the following marking scheme :**
- Full Mark* : +4 for **CORRECT ANSWER**;
Zero Marks : 0. If none of the option is chosen (i.e. the question is unanswered)
-1 For **INCORRECT ANSWER**

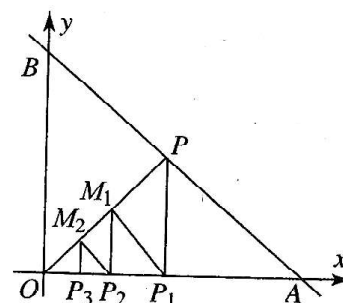
MATHEMATICS

SECTION - 1

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.20 has four choices (1), (2), (3), (4) out of which **ONLY ONE** is correct.

1. If a, b, c, d are in G.P., then $(b - c)^2 + (c - a)^2 + (d - b)^2$ is equal to
 (1) $(a - d)^2$ (2) $(ad)^2$ (3) $(a + d)^2$ (4) $(a/d)^2$
2. If the line $y = \sqrt{3}x$ cuts the curve $x^3 + y^3 + 3xy + 5x^2 + 3y^2 + 4x + 5y - 1 = 0$ at the points A, B, C then OA. OB. OC is :
 (1) $\frac{4}{13}(3\sqrt{3} - 1)$ (2) $3\sqrt{3} + 1$ (3) $\frac{2}{\sqrt{3}} + 7$ (4) none of these
3. If $al + bm + cn = 0$, then the family of lines $lx + my + n = 0$ are concurrent at :
 (1) $\left(\frac{a}{c}, \frac{b}{c}\right)$ (2) $\left(\frac{b}{c}, \frac{a}{c}\right)$ (3) $\left(\frac{a}{b}, \frac{c}{b}\right)$ (4) none of these
4. The internal angle bisector of angle A in the triangle A $(-8, 5)$, B $(-15, -19)$ and C $(1, -7)$ is $ax + 2y + c = 0$, then the values of a and c are respectively :
 (1) $(11, 78)$ (2) $(78, 11)$ (3) $(11, 5)$ (4) None of these
5. The distance of point $(2, 3)$ from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to
 (1) $4\sqrt{2}$ (2) $6\sqrt{3}$ (3) $3 + 4\sqrt{2}$ (4) $4 + 6\sqrt{3}$
6. A light ray emits from the origin making an angle 30° with the positive x -axis. After getting reflected by the line $x + y = 1$, if this ray intersects x -axis at Q, then the abscissa of Q is :
 (1) $\frac{2}{\sqrt{3} - 1}$ (2) $\frac{2}{3 + \sqrt{3}}$ (3) $\frac{2}{3 - \sqrt{3}}$ (4) $\frac{\sqrt{3}}{2(\sqrt{3} + 1)}$
7. For $t \in (0, 2\pi)$, If ABC is an equilateral triangle with vertices A $(\sin t, -\cos t)$, B $(\cos t, \sin t)$ and C (a, b) such that its orthocentre lies on a circle with centre $\left(1, \frac{1}{3}\right)$, then $(a^2 - b^2)$ is equal to :
 (1) $\frac{8}{3}$ (2) 8 (3) $\frac{77}{9}$ (4) $\frac{80}{9}$
8. The line $x + y = 1$ meets the x -axis at A and the y -axis at B, P is the mid-point of AB; P_1 is the foot of the perpendicular from P to OA; M_1 is that P_1 from OP; P_2 is that of M_1 from OA; M_2 is that of P_2 from OP; P_3 is that of M_2 from OA; and so on. If P_n denotes the n th foot of the perpendicular on OA, then OP_n is
 (1) $\left(\frac{1}{2}\right)^{n-1}$ (2) $\left(\frac{1}{2}\right)^n$ (3) $\left(\frac{1}{2}\right)^{n+1}$ (4) none of these



9. Find the equation of the angle bisector of the angle formed by the lines $12x + 5y - 10 = 0$ and $3x + 4y - 5 = 0$ containing the point $(1, 1)$.
 (1) $7x - 9y + 5 = 0$ (2) $7x - 9y + 8 = 0$ (3) $9x + 7y + 5 = 0$ (4) $9x + 7y + 8 = 0$
10. Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that $\triangle ABC$ is an equilateral triangle. Then, the area of the $\triangle ABC$ is:
 (1) $3\sqrt{3}$ (2) $2\sqrt{3}$ (3) $\frac{8}{\sqrt{3}}$ (4) $\frac{10}{\sqrt{3}}$
11. The shortest distance between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 10x - 10y + 41 = 0$ is :
 (1) $\sqrt{4} - 1$ (2) 0 (3) $\sqrt{4}$ (4) $5\sqrt{2} - 4$
12. Two circles touch the x – axis and the line $y = mx$. They meet at $(9, 6)$ and at one more point and the product of their radii is $\frac{117}{2}$, then the value of m is :
 (1) $2\sqrt{2}$ (2) $\sqrt{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) none of these
13. A circle C_1 is drawn having any point P on x – axis as its centre and passing through the centre of the circle (C) $x^2 + y^2 = 1$. A common tangents to C_1 and C touch the circles at Q and R respectively. Then Q (x , y) always satisfies :
 (1) $x^2 - 1 = 0$ (2) $x^2 + y^2 = 1$ (3) $y^2 - 1 = 0$ (4) none of these
14. Which of the following is true?
 (1) The image of the circle $x^2 + y^2 = 1$ in the line $x + y = 2$ is $x^2 + y^2 + 4x - 4y + 7 = 0$
 (2) The maximum distance of the point $(4, 4)$ from the circle $x^2 + y^2 - 2x - 15 = 0$ is 10
 (3) The image of the circle $x^2 + y^2 = 1$ in the line $x + y = 2$ is $x^2 + y^2 + 4x + 4y + 7 = 0$
 (4) If two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ cut the co – ordinate axes in concyclic points then $a_1a_2 = b_1b_2$
15. The value (s) of m for which the line $y = mx$ lies wholly outside the circle $x^2 + y^2 - 2x - 4y + 1 = 0$, is (are):
 (1) $m \in (-4/3, 0)$ (2) $m \in (-4/3, 0]$ (3) $m \in (0, 4/3)$ (4) $m \in (-2/3, 1]$
16. If $c + 1 < b$, then roots of $x^2 - bx + c = 0$;
 (1) are real and equal
 (2) are real and distinct
 (3) are imaginary
 (4) noting can be said about the nature of roots as given condition is not sufficient
17. If x is real, then $x/(x^2 - 5x + 9)$ lies in
 (1) $\left[-1, \frac{-1}{11}\right]$ (2) $\left[\frac{-1}{11}, 1\right]$ (3) $\left[\frac{1}{11}, 1\right]$ (4) none of these
18. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$
 (1) 0 (2) 8000 (3) 8080 (4) 16000

19. Given two series $S_1 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{1001}$ and $S_2 = \frac{1}{501} + \frac{1}{502} + \frac{1}{503} + \dots + \frac{1}{1001}$, then :
- (1) $S_1 > S_2$ (2) $S_1 < S_2$ (3) $S_1 = S_2$ (4) $S_1 + S_2 = 501$
20. $\sum_{r=1}^{\infty} \frac{r^3 + (r^2 + 1)^2}{(r^4 + r^2 + 1)(r^2 + r)}$ is equal to :
- (1) $\frac{3}{2}$ (2) 1 (3) 2 (4) infinite

SECTION - 2 [NUMERICAL TYPE]

Q.21 to Q.25 are “Numerical Type” questions.

21. The equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ have four positive real roots. Then $a + b$ is equal to
22. If (α, β) is the orthocentre of the triangle ABC with vertices A(3, -7), B(-1, 2) and C(4, 5), then $9\alpha - 6\beta + 60$ is equal to :
23. Let $S = \frac{4}{19} + \frac{44}{19^2} + \frac{444}{19^3} + \dots$ up to ∞ . Then the value of $81S$ is equal to
24. Two circles are given as $x^2 + y^2 + 14x - 6y + 40 = 0$ and $x^2 + y^2 - 2x + 6y + 7 = 0$ with their centres as C_1 and C_2 . If equation of another circle whose centre C_3 lies on the line $3x + 4y - 16 = 0$ and touches the circle C_1 externally and also $C_1C_2 + C_2C_3 + C_3C_1$ is minimum is $x^2 + y^2 + ax + by + c = 0$ then the value of $(a + b + c)$ is
25. Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^3b^3c^2d$ is 3750β , then the value of β is :

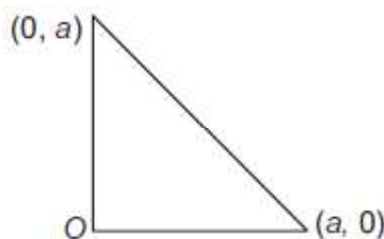
PHYSICS

SECTION - 1

[SINGLE CORRECT CHOICE TYPE]

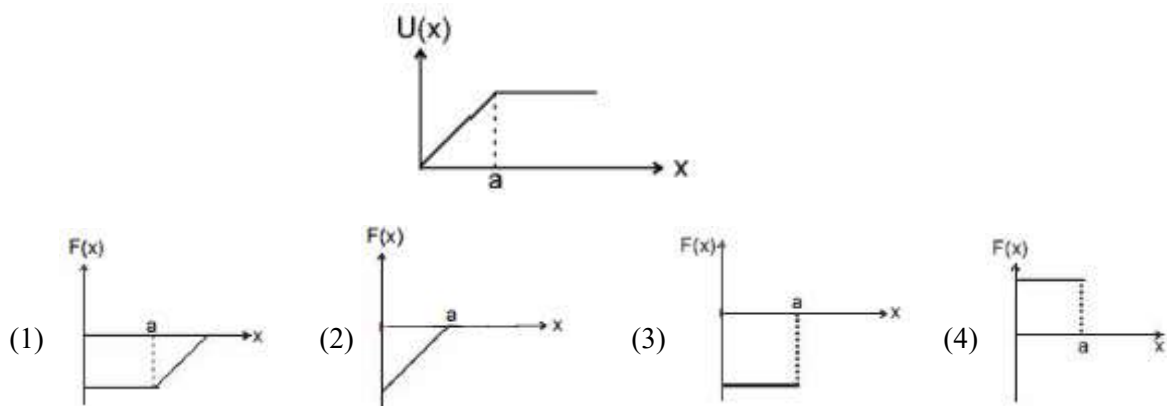
Q.26 to Q.45 has four choices (1), (2), (3), (4) out of which **ONLY ONE** is correct.

26. Three rods of the same mass are placed as shown in figure. What will be the coordinates of centre of mass of the system?

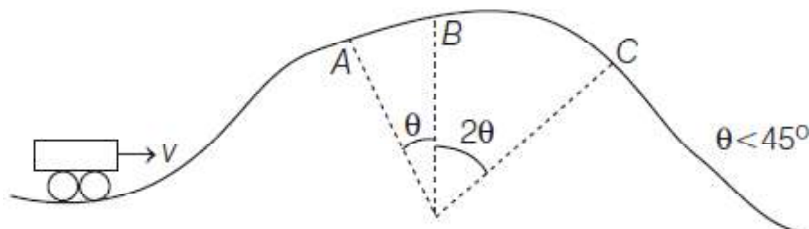


- (1) $\left(\frac{a}{2}, \frac{a}{2}\right)$ (2) $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ (3) $(\sqrt{2}a, \sqrt{2}a)$ (4) $\left(\frac{a}{3}, \frac{a}{3}\right)$

27. Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between $t = 2\text{ s}$ and $t = 4\text{ s}$.
 (1) 32 J (2) 46 J (3) 61 J (4) 100 J
28. At time $t = 0\text{ s}$ a particle starts moving along the x -axis. If its kinetic energy increases uniformly with time ' t ', the net force acting on it must be proportional to:
 (1) constant (2) t (3) $\frac{1}{\sqrt{t}}$ (4) $\sqrt{t}$
29. The potential energy of a system is represented in the first figure, the force acting on the system will be represented by:

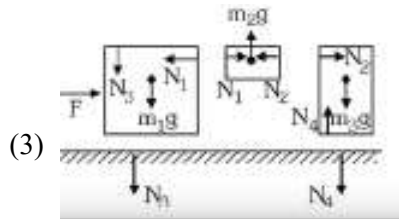
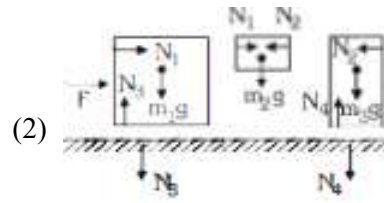
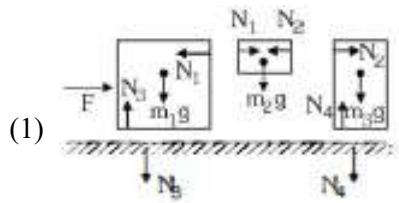
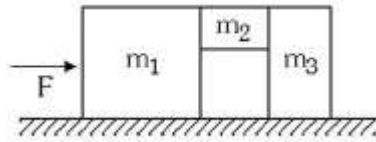


30. A car of mass m starts moving so that its velocity varies according to the law $v = \beta\sqrt{s}$, where β is a constant, and s is the distance covered. The total work performed by all the forces which are acting on the car during the first t seconds after the beginning of motion is:
 (1) $m\beta^4 t^2 / 8$ (2) $m\beta^2 t^4 / 8$ (3) $m\beta^4 t^2 / 4$ (4) $m\beta^2 t^4 / 4$
31. A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table?
 (1) 7.2 J (2) 3.6 J (3) 120 J (4) 1200 J
32. A self-propelled vehicle (assume it as a point mass) runs on a track with constant speed v . It passes through three positions A, B and C on the **circular part** of the track. Suppose N_A , N_B and N_C are the normal forces exerted by the track on the vehicle when it is passing through points A, B and C respectively, then :



- (1) $N_A = N_B = N_C$ (2) $N_B > N_A > N_C$ (3) $N_C > N_A > N_B$ (4) $N_B > N_C > N_A$

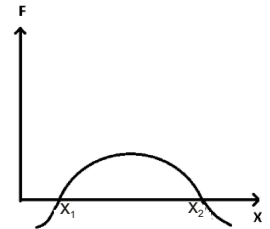
33. All surface smooth, Draw F.B.D :



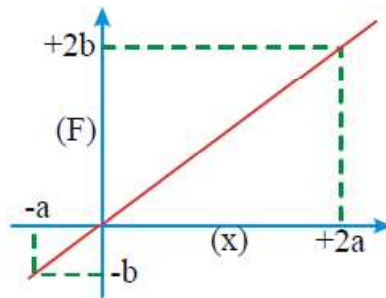
(4) None of these

34. The force acting on a body moving along x axis varies with the position of the particle as shown in the figure. The body is in stable equilibrium at :

- (1) $x = x_1$ (2) $x = x_2$
 (3) Both x_1 and x_2 (4) Neither x_1 nor x_2

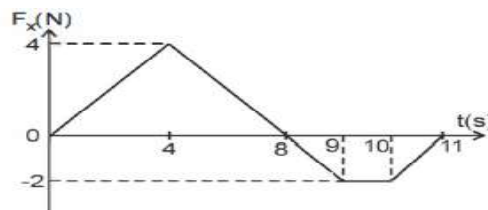


35. A force F acting on a particle varies with the position x as shown in the graph. Find the work done by the force in displacing the particle from $x = -a$ to $x = +2a$:



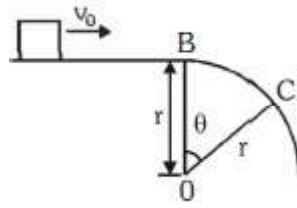
- (1) $\frac{3ab}{2}$ (2) $\frac{4ab}{2}$ (3) $\frac{2}{3ab}$ (4) $\frac{2}{4ab}$

36. A 1 kg toy car can move along an x axis. Graph shows force F_x , acting on the car which starts from rest at time $t = 0$. The kinetic energy of particle at $t = 10$ s is :



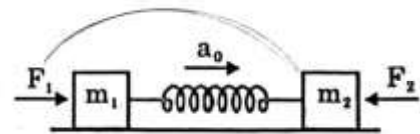
- (1) 100 J (2) 169 J (3) 84.5 J (4) None of these

37. A small block slides with velocity $0.5\sqrt{gr}$ on the horizontal frictionless surface as shown in the figure. the block leaves the surface at point C. The angle θ in the figure is :



- (1) $\cos^{-1}(4/9)$ (2) $\cos^{-1}(3/4)$ (3) $\cos^{-1}(1/2)$ (4) None of the above
38. Two blocks m_1 and m_2 are connected with a compressed spring and placed on a smooth horizontal surface as shown in figure. Force constant of spring of k . Under the influence of forces F_1 and F_2 , at an instant blocks move with common acceleration a_0 . At that instant F_2 is suddenly withdrawn. Mark **INCORRECT** option.:

(1) Instantaneous acceleration of m_1 is $a_0 - \frac{F_1}{m_1}$



(2) Instantaneous acceleration of m_2 is $a_2 = a_0 + \frac{F_2}{m_2}$

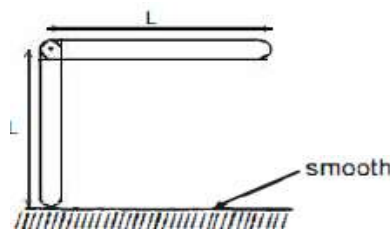
(3) Instantaneous acceleration of m_1 is $a_1 = a_0$

(4) Spring force is $F_{\text{spring}} = m_2 a_0 + F_2$

39. A particle of mass $4m$ which is at rest explodes into masses m , m & $2m$. Two of the fragments of masses m and $2m$ are found to move with equal speeds v each in opposite directions. The total mechanical energy released in the process of explosion is:

(1) mv^2 (2) $2mv^2$ (3) $1/2 mv^2$ (4) $4 mv^2$

40. Two identical rods are joined at one of their ends by a pin. Joint is smooth and rods are free to rotate about the joint. Rods are released in vertical plane on a smooth surface as shown in the figure. The displacement of the joint from its initial position to the final position is (i.e. when the rods lie straight on the ground):



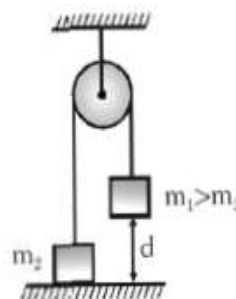
(1) $\frac{L}{4}$ (2) $\frac{\sqrt{17}}{4}L$ (3) $\frac{\sqrt{5}}{2}L$ (4) none of these

41. A body kept on a smooth inclined plane inclination 1 in x will remain stationary relative to the inclined plane if the plane is given a horizontal acceleration equal to :-

(1) $\sqrt{x^2 - 1}g$ (2) $\frac{\sqrt{x^2 - 1}}{x}g$ (3) $\frac{gx}{\sqrt{x^2 - 1}}$ (4) $\frac{g}{\sqrt{x^2 - 1}}$

42. If masses are released from the position shown in figure then time elapsed before mass m_1 collides with the floor will be:

(1) $\sqrt{\frac{2m_1gd}{m_1 + m_2}}$ (2) $\sqrt{\frac{2(m_1 + m_2)d}{(m_1 - m_2)g}}$
 (3) $\sqrt{\frac{2(m_1 - m_2)d}{(m_1 + m_2)g}}$ (4) None of these



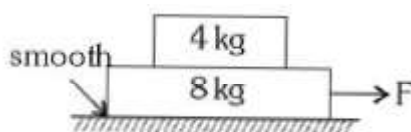
43. A solid of mass m is whirled in a vertical circle with constant speed v by attaching it with a string. The difference in tension in the string at lowest point and highest point is:

(1) mg (2) $2mg$ (3) $\frac{2mv^2}{r}$ (4) zero

44. A heavy particle hangs from a point O , by a string of length a ; It is projected horizontally with a velocity $v = \sqrt{(2 + \sqrt{3})ag}$. The angle with the downward vertical, string makes where string becomes slack is

(1) $\theta = \sin^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ (2) $\theta = \cos^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ (3) $\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (4) $\theta = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

45. A block of 4 kg is placed on a plank having mass 8 kg. A force $F = 20$ N is applied on plank. Then find the friction force between 4 kg and 8 kg block. Here coefficient of friction between 4 kg and 8 kg is $\mu = 0.4$, floor is smooth :



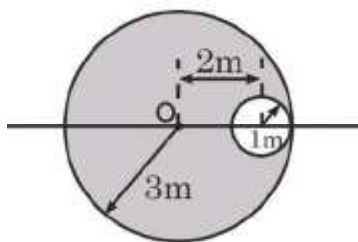
(1) 16 N (2) $\frac{20}{3}$ N (3) 32 N (4) None of these

SECTION - 2

[NUMERICAL TYPE]

Q.46 to Q.50 are “Numerical Type” questions.

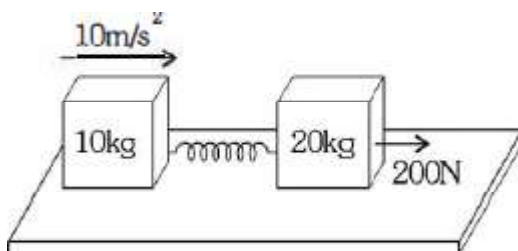
46. A ball of mass 4 kg is moving with a velocity of 10 ms^{-1} , collides with a spring of length 8 m and force constant 100 Nm^{-1} . The length of the compressed spring is x m. The value of x , to the nearest integer, is
47. A weightless thread can bear tension upto 3.7 kg-wt. A stone of mass 500 g is tied to it and revolved in a circular path of radius 4 m in a vertical plane. If $g = 10 \text{ ms}^{-2}$, then the maximum angular velocity (in rad/s) of the stone will be
48. In the figure shown find out the distance of centre of mass (in cm) of a system of a uniform circular plate of radius 3m from O in which a hole of radius 1m is cut whose centre is at 2m distance from centre of large circular plate.



49. One end of a massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N. with what value of minimum safe acceleration (in ms^{-2}) can a monkey of 60 kg move down on the rope



50. The masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass. At the instant shown, the 10 kg mass has acceleration 10 m/sec^2 then calculate the acceleration of 20 kg mass in m/s^2



CHEMISTRY

SECTION - 1

[SINGLE CORRECT CHOICE TYPE]

Q.51 to Q.70 has four choices (1), (2), (3), (4) out of which **ONLY ONE** is correct.

51. Under identical conditions of pressure and temperature, 4L of gaseous mixture (H_2 and CH_4) effuses through a pin hole for 5 mins whereas 4L of a gas X of molecular mass 36 amu takes 10 mins to effuse through the same pin hole. The mole ratio of $H_2 : CH_4$ in the initial mixture is-
(1) 1 : 2 (2) 2 : 1 (3) 2 : 3 (4) 1 : 1
52. On increasing the temperature of an ideal gas at constant volume, the mean free path-
(1) Increases
(2) Decreases
(3) remains unchanged
(4) May or may not change depending on the size of gas molecules
53. Which of the following is correct expression for potential energy for an e^- in Bohr's orbit.
(1) $-\frac{KZe^2}{2r}$ (2) $-\frac{KZe^2}{r^2}$ (3) $-\frac{Ze^2}{r}$ (in C.G.S.) (4) $-\frac{Ze^2}{2r}$ (in C.G.S.)
54. The difference between wave number of 1st line of Lyman series & last line of Paschen series for hydrogen is (R_H = Rydberg constant)
(1) $\frac{R_H}{36}$ (2) $\frac{23R_H}{36}$ (3) $\frac{9R_H}{36}$ (4) $\frac{R_H}{4}$
55. In a hydrogen like sample electron is in 2nd excited state, the energy of 4th state of this sample is -13.6 eV, then incorrect statement is:
(1) Atomic number of element is 4
(2) 3 different types of spectral line will be observed if electrons make transition upto ground state from the 2nd excited state.
(3) 2nd line of Balmer series of this sample has same energy value as 1st excitation energy of H-atoms.
(4) A 25 eV photon can set free the electron from the 2nd excited state of this sample.
56. Which of the following about SF_4 , SOF_4 , CH_2SF_4 and OCF_2 molecules is correct-
(1) Equatorial FSF bond angle in SOF_4 will be less than that in SF_4 molecule
(2) The bond angle FCO will be $< 120^\circ$ in molecule OCF_2 .
(3) The two hydrogens, carbon, sulphur and two fluorines (of axial positions) in molecule CH_2SF_4 will be lying in the same plane.
(4) The axial FSF bond angle in $SF_4 = 180^\circ$
57. The distance of spherical nodes from nucleus for the given orbital are

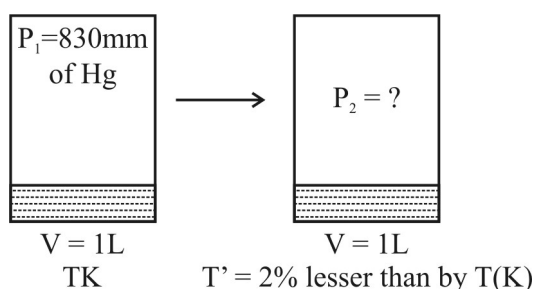
$$\Psi_{\text{radial}} = \frac{1}{9\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} [(\sigma^2 - 4\sigma + 3)] e^{(-\sigma/2)} \text{ where } a_0 \text{ and } Z \text{ are the constants and } \sigma = \frac{2Zr}{a_0}$$

- (1) zero, infinity (2) $\frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$ (3) $\frac{a_0}{Z}, \frac{3}{2} \frac{a_0}{Z}$ (4) $\frac{3}{2} \frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$

58. Match the following;

- | | |
|------------------------|--------------------------|
| (P) XeF_5^+ | (1) Tetrahedral |
| (Q) SiF_5^- | (2) Square planar |
| (R) AsF_5^+ | (3) Trigonal bipyramidal |
| (S) ICl_4^- | (4) Square pyramidal |
| | (5) Octahedral |
| (1) P-1, Q-2, R-5, S-3 | (2) P-1, Q-3, R-5, S-2 |
| (3) P-4, Q-2, R-1, S-3 | (4) P-4, Q-3, R-1, S-2 |

59. A jar contains a gas and some amount of water at temperature 'T' K. The total pressure in the jar is 830 mm of Hg. Calculate final total pressure (in mm of Hg) in the jar if the temperature of jar is reduced by 2%. (Assume volume occupied by liquid water is negligible).



(The aqueous tension of water at the given temperature T and T' are 30 mm of Hg and 25 mm of Hg respectively).

- (1) 71 mm Hg (2) 809 mm Hg (3) 825 mm Hg (4) 900 mm Hg
60. How many of the following oxy acids does not contain E—O—E bond (E = central element)
 $\text{H}_2\text{S}_2\text{O}_7$, $\text{H}_2\text{S}_2\text{O}_8$, $\text{H}_2\text{S}_2\text{O}_3$, $\text{H}_2\text{S}_2\text{O}_5$, $\text{H}_2\text{S}_2\text{O}_4$, $\text{H}_4\text{P}_2\text{O}_6$, $\text{H}_4\text{P}_2\text{O}_7$, $\text{H}_2\text{N}_2\text{O}_2$, $\text{H}_2\text{B}_4\text{O}_7$
 (1) 2 (2) 3 (3) 4 (4) 6
61. Arrange the following compounds in order of increasing dipole moment:
 (I) 1,3,5-Trichloro benzene
 (II) 1,2,4-Trichloro benzene
 (III) 1,2,3,4-Tetra chlorobenzene
 (IV) P-dichloro benzene
 (1) $\text{I} = \text{IV} < \text{II} < \text{III}$ (2) $\text{IV} < \text{I} < \text{II} < \text{III}$
 (3) $\text{IV} = \text{I} < \text{III} < \text{II}$ (4) $\text{IV} < \text{II} < \text{I} < \text{III}$
62. In which of the following pairs hybridisation of the central atom are different?
 (1) ClF_3 , ClF_3O (2) ClF_3O , ClF_3O_2
 (3) $[\text{ClO}_4\text{O}]^-$, $[\text{XeOF}_4]$ (4) $[\text{ClF}_2\text{O}]^+$, $[\text{ClF}_4\text{O}]^-$
63. The structure of F_2SeO is analogous to:
 (1) SO_3 (2) ClO_3^- (3) XeO_3 (4) both (2) and (3)
64. Phosphorous penta-chloride in gaseous phase exist as a monomer. In solid state, it exist as PCl_4^+ and PCl_6^- ions. The hybrid state of P-atom in PCl_5 is sp^3d . The hybrid states of P-atoms in PCl_4^+ and PCl_6^- will be:
 (1) sp^3d , sp^3d^2 (2) sp^3 , sp^3d^2 (3) sp^3d^2 , sp^3d (4) sp^3 , sp^3d

65. The resultant dipole moment of water is 1.85 D ignoring the effect of lone pair. The dipole moment of each OH bond will be
(Given : bond angle in $\text{H}_2\text{O} = 104^\circ$, $\cos 104^\circ = -0.25$)
(1) 1.51 D (2) 1.32 D (3) 1.42 D (4) 1.63 D
66. The root mean square velocity of one mole of a monatomic gas having molar mass M is U_{rms} . The relation between the average kinetic energy (E) of the gas and U_{rms} is:
(1) $U_{\text{rms}} = \sqrt{\frac{3E}{2M}}$ (2) $U_{\text{rms}} = \sqrt{\frac{2E}{3M}}$ (3) $U_{\text{rms}} = \sqrt{\frac{2E}{M}}$ (4) $U_{\text{rms}} = \sqrt{\frac{E}{3M}}$
67. At the same temperature and pressure, which of the following gas will have the highest kinetic energy per mole.
(1) Hydrogen (2) Oxygen
(3) Methane (4) All will have the same kinetic energy
68. At what temperature, the sample of neon gas would be heated to double of its pressure, if the initial volume of gas is reduced to 15% at 75°C .
(Report nearest integer)
(1) 319°C (2) 592°C (3) 128°C (4) 60°C
69. A box of 1L capacity is divided into two equal compartments by a thin partition which are filled with 2g H_2 and 16g CH_4 respectively. The pressure in each compartment is recorded as 5 atm. The total pressure when partition is removed will be:
(1) 5 atm (2) 10 atm (3) 2.5 atm (4) 1.25 atm
70. Compressibility factor of methane gas at 300 K and 1 atm is 0.8. The mass of methane gas present at 1 atm and 300 K in 1 L cylinder is nearly :
[Given : $R = 0.08 \text{ lit atm mol}^{-1} \text{ K}^{-1}$].
(1) 1.08 gm (2) 0.75 gm (3) 0.83 gm (4) 0.93 gm

SECTION - 2

[NUMERICAL TYPE]

Q.71 to Q.75 are “Numerical Type” questions.

71. If a photon of wavelength 310 nm falls on a metal surface having work function 3eV then the maximum KE of photoelectron will be: (in eV)
72. Find total number of polar molecules among the following.
 PF_3Cl_2 , SF_4 , PCl_5 , PCl_3F_2 , SF_6 , XeF_2 , NO_2^+ , BF_2Cl , BF_3
73. Consider y-axis as internuclear axis, how many of following will lead to π bond formation:
(i) $p_y - p_y$ (ii) $p_x - p_x$ (iii) $p_z - p_z$ (iv) $d_{xy} - d_{xy}$
(v) $d_{yz} - d_{yz}$ (vi) $p_x - d_{xy}$ (vii) $d_{xy} - p_z$ (viii) $d_{xz} - d_{xz}$
74. A mixture of N_2 and H_2 has initially mass ratio of 196 : 1 then find after how many steps we can obtain a mixture containing 1 : 14 mole ratio of N_2 and H_2 .
75. A 0.5 dm^3 flask contains gas A and 1 dm^3 flask contains gas B at the same temperature. If density of A = 3 g/dm^3 and that of B = 1.5 g/dm^3 and the molar mass of A = $1/2$ of B, the ratio of pressure exerted by gases is:

* * * * *

Answer Key

Grade : XI-JEE

Review Test-5 (MAIN)

Date : 22-08-2026

CODE : 20105

MATHEMATICS																								
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
1	1	1	1	4	2	2	2	1	3	4	1	1	4	1	2	2	4	3	1	2	25	38	0	90
PHYSICS																								
26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
4	2	3	3	1	2	2	1	2	1	3	2	1	2	2	4	2	2	2	2	6	4	25	4	5
CHEMISTRY																								
51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
4	3	3	2	3	3	4	4	2	2	1	4	4	2	1	3	4	1	1	3	1	3	5	4	4